



The Interaction of Control Systems and Stakeholder Networks in Shaping the Identities of Self-Managed Teams

Organization Studies

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Abstract

Team identity has received little research attention even though an increasing number of firms are moving to team-based organizations and there is evidence that teams form identities. We explore the extent to which team identity can be institutionalized as a central organizing principle of team-based firms. We argue that managerial and stakeholder interventions shape the self-construction of team identity as well as the team's commitment to specific work objectives. We also suggest that team identity becomes isomorphic to organizational identity because of pressures related to: (1) the presence of a dense network of managers and stakeholders, which orients teams towards a focus on certain aspects of the higher-order identity; (2) the use of team routines and regular feedback loops, which force alignment with the organizational identity; and (3) the use of coordinating roles aimed at promoting, ratifying and reinforcing the convergence of identity within the team. We analyse multiple cases from a major multinational corporation in the telecommunications industry, which we examine through the lens of a multi-level model of controls involving the micro, meso and macro organizational levels. We expand and refine the model in the process.

Keywords

managerial control, organizational control, organizational identity, team identity

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Introduction

An understanding of 'identity' is important for comprehending the functioning of those contemporary organizations that rely on flat hierarchies and the use of more or less autonomous teams (Albert, Ashforth, & Dutton, 2000). Thus far, research has paid little attention to the identity aspects of these new organizational forms. Identity is important because of its powerful sense-making, direction-setting and motivating roles, and there is no reason identity should be any less important in these regards in contemporary, team-based organizations. In this study, we apply an identity perspective to a particular instance of contemporary work organization, namely 'agile' software development (Martin, 2003).

Agile methods have been adopted in many industries (Austin & Devin, 2009). They represent a new model of labour-process control that maintains discipline in teamwork. Post-bureaucratic control relies on the creation of shared meaning, which reduces the need for the 'principles of hierarchy and explicitly rule-governed behavior' (Sewell, 1998, p. 408). An internalized cognitive structure of the organization and its goals (Albert et al., 2000) serves as the basis of new control strategies designed to coexist with established bureaucratic frameworks. This results in a 'complex interweaving between bureaucratic logic and the autonomisation of conduct' (Fournier, 1999, p. 293).

Identity is pivotal to our understanding of coordination in organizations. Organizational identity is the cognitive image held by individuals within the organization that is used to make sense of the world (Dutton, Dukerich, & Harquail, 1994). Identity provides rules of action that help organizational actors deal with ambiguity and cognitive limitations by emphasizing certain issues and problems. It also helps them define those issues that are crucial and demand attention (Thornton, 2002). The extant research indicates that the identities of organizational members are closely connected with organizational identity (Ashforth & Mael, 1989; Brown, 1997; Dutton et al., 1994; Elsbach & Kramer, 1996). However, to the best of our knowledge, the extant research does not theorize about the mechanisms that link team identity and organization identity. Nevertheless, the literature emphasizes the importance of observing cross-level dynamics and theoretically unfolding the 'cascades' of nested identities, such as the link between the organizational and team levels (Ashforth, Rogers, & Corley, 2011). Identities are not created in a vacuum and organizational identity construction is not performed by an abstract organizational entity. Rather, it is crafted by managers and stakeholders who are simultaneously creating their own identities through self-categorization (Kramer, 1991), cognitive identification (Dutton et al., 1994) and self-affirmation (Brown, 1997; Elsbach & Kramer, 1996).

We build on the idea that 'organizational identity is best understood as contested and negotiated through iterative interactions between managers and stakeholders' (Scott & Lane, 2000, p. 44). Scott and Lane (2000) suggest a number of relations between stakeholder characteristics, such as the density of their related networks and stakeholders' abilities to influence managers' reconstruction of organizational identity. We expand on these ideas, especially on the view that organizational identity is an outcome of the complex interactions among managers, organizational members and other stakeholders. In this regard, we argue that team members are not passive targets of managerially designed identities. We consider several groups of organizational actors and constituencies at multiple organizational levels as we integrate theory on organizational identity with theory on team identification. We apply a multi-level model of controls (Martin, 2003) at three organizational levels: the micro or team level, the meso level and the macro level.

Based on a field investigation of three R&D units in the same multinational telecommunications company, we argue that agile routines make teams permeable to stakeholder influence at the meso level. At the same time, organizational routines at the macro level enable intense relations among stakeholders, leading to the formation of coalitions that pressure teams. This pushes teams towards

conformity with stakeholders' values and beliefs, thereby prompting alignment between the identities of teams and those of stakeholders. In addition, we propose that task interdependencies within the team at the micro level intensify the strength of relational ties among members. As team connectivity becomes more intense, team members tend to behave in similar ways despite their freedom to be different, which gives rise to a set of legitimized working conventions (Scott, 1995) nested in value-based control. This team identity becomes the basis on which members operate when developing products or services. Finally, we propose that established managerial controls at the macro level tend to amplify the defined team identity, adding legitimacy to team behaviours as the team works towards agile values and principles instilled by stakeholders.

Theoretical Background

Organizational stakeholders in the context of organizational identity construction

In general, stakeholders are groups or individuals with relationships with a firm that have the power to affect the firm's performance and/or have a stake in the firm's performance (Freeman, 1983). In this paper, 'stakeholders' refers to individuals outside the focal team who have an interest in leading the team towards the desired performance. Such stakeholders might include middle managers, line managers, customer representatives and team coordinators, who act as dominant, powerful and legitimate stakeholders (Mitchell, Agle, & Wood, 1997) and who attract management's attention.

In modern organizations, management may use social identity and 'the corporatization of the self' (Kärreman & Alvesson, 2004, p. 151) as a way to exercise meaningful executive authority over other social groups. For this purpose, management uses socio-ideological control in attempts to control worker beliefs and technocratic control in attempts to control worker behaviour (Kärreman & Alvesson, 2004). Given their need to be sensitive to their obligations to stakeholders (Clarkson, 1995), to serve the strategic goals of the organization and to satisfy their own needs of self-definition, managers may initiate interactions with stakeholders and other organizational entities through which organization identity can be inferred and controlled.

In the process of constructing organizational identity, top managers play a special role with respect to stakeholders, as they can be considered 'agents of [all] other stakeholders' (Hill & Jones, 1992, p. 135). Previous studies have focused on the roles of top management (Gioia & Thomas, 1996) and other organizational members (Elsbach & Kramer, 1996) in organizational identity construction. We define organizational identity as the shared beliefs held by organizational members concerning the main, enduring and distinguishing peculiarities of the company (Albert & Whetten, 1985). Organizational identity construction is dynamic and iterative, and it results from managers' and stakeholders' consideration of the meaning of events and policies, and their interpretations of identity-relevant feedback (Scott & Lane, 2000).

Organizational identity construction and managerial organizational controls

Managers and stakeholders exercise control and power in the organization in order to secure sufficient resources and steer individual actions towards specific ends (Langfield-Smith, 1997). Control strategies may contribute to identity formation, but the multiple identity demands may create tensions and even challenge the 'very core of each employee's sense of selfhood and identity, defining his/her very being' (Gabriel, 1999, p. 180). Managerial control typically involves mechanisms for monitoring and evaluating the inputs and/or outputs of relevant organizational entities, or mechanisms that target employee beliefs.

Two layers of control are considered in this study: technocratic and socio-ideological (Alvesson & Kärreman, 2001). Specifically, technocratic control encompasses the use of bureaucracy and performance measures. As such, it is associated with structural arrangements (e.g. organizational hierarchies, plans, feedback procedures, systems) that are focused on the behaviours and work methodologies that make work processes as simple and transparent as possible. Socio-ideological control, on the other hand, targets beliefs via norms, emotions and values, and it indirectly affects the behaviour of workers. In ideological forms of control, managers seek to make employees comply with their own values and ideals.

Research suggests that there has been a general and significant shift in managerial control systems in complex, decentralized organizations from technocratic control towards socio-ideological control (Alvesson & Kärreman, 2001). Technocratic control is oriented towards the behavioural level, while bureaucracy may have a socio-ideological influence. The two forms of control 'feed upon and inform each other' (Alvesson & Kärreman, 2001, p. 171). In the following, we investigate key mechanisms that are involved in this interplay.

Team identity construction within nested organizational identities

Organizational identities are linked across levels of analysis: action and structure can both constrain (by restricting choice) and favour (by enabling choice within those boundaries) each other (Bechky, 2011; Giddens, 1984). Thus, the macro and micro levels of analysis are mutually linked in the sense that identities at higher levels of analysis both constrain and enable the formation and enactment of identities at lower levels, which then feed back into the formation and enactment of higher-order identities.

In terms of cross-level effects, a given collective identity will have the greatest impact on identities at proximal (especially adjacent) levels, with ripple effects possibly affecting other levels. After lower-order and higher-order identities are established, the downward effects may be stronger than the upward effects due to the fact the power is generally closely correlated with hierarchical level (Pfeffer, 1981). While organizational identity is best understood as contested and negotiated through iterative interactions between managers and stakeholders, we argue that the construction of team identity also involves teams in iterative interactions with managers and stakeholders. Therefore, we suggest that team identity is crafted by interactions between 'insiders' and 'outsiders' (Coupland & Brown, 2004) in a network of social relationships. As a team's survival is contingent on its compliance with expectations from stakeholders and managers, an analysis of this network of social relationships may help determine the degree of resistance within a team to the higher-level organizational identity. Resistance arises from over-identification mechanisms (Dukerich, Kramer, & Parks, 1998) and from the desire to preserve the team's sense of self, which make team members devoted to the known value system and reluctant to change (Ferraris, Carveth, & Parrish-Sprowl, 1993).

Individual identity construction within the context of team identity construction

Research defines identification using social identity and social categorization (Turner, 1982) theories, and points to the cognitive links between individuals and groups that emerge from the images or mental representations that members have of their group. The desired identity image (Schlenker, 1985) (i.e. self-images that preserve or enhance individual self-esteem; Leary, 1983) is built by making an initial assessment of the audience (e.g. expectations, goals, beliefs) and the related situational characteristics (e.g. social roles, rules) within the social context in which individuals operate. Thus, individuals modify the desired image based on their perceptions of the audience's needs,

beliefs, knowledge and skills (Schlenker & Leary, 1982). This audience includes the self, reference groups and interaction partners (Schlenker, 1985).

Cognitive identification with a group has a strong impact on members' behaviour and, consequently, on group outcomes (Ashforth & Mael, 1989; Dutton et al., 1994). When members socially identify with a group, they embrace its main problems as their own and collaborate in their resolution, regardless of their feelings toward other group members (Brewer & Gardner, 1996). In addition, self-categorization as a member emphasizes one's self-perception of similarity to the group prototype (Hogg & Terry, 2000) and thereby enacts the creation of a group identity. In this sense, the depersonalization of the self 'is the process underlying group phenomena' (Hogg & Terry, 2000, p. 123). In other words, group dynamics cannot be analysed separate from the categorization of group members and their identity processes. However, some researchers (Albert & Whetten, 1985; Ashforth & Mael, 1989) show that group identity is shaped by the characteristics of the group, such as its goals, mission, practices and actions, and is not related to its individual members (Albert & Whetten, 1985; Ashforth & Mael, 1989).

Research gaps

The construction of organizational identities is a complex phenomenon. Although the literature notes that organizational members' identities and organizational identity are closely connected (Ashforth & Mael, 1989; Brown, 1997; Dutton et al., 1994; Elsbach & Kramer, 1996), little attention has been devoted to identity construction mechanisms or to the relationships among them. Most of the extant research focuses on a particular level of analysis, typically the macro level (Ashforth et al., 2011). As such, it neglects (complex) inter-level dynamics in the construction of identity (for exceptions, see Cheney & Christensen, 2000; Gioia, Schultz, & Corley, 2000).

Alvesson and Willmott (2002) present identity regulation as a form of managerial organization control and offer a preliminary overview of how identity can be influenced, regulated and changed within organizations. Their research provides a simplified view of identity change, as it does not focus on the cross-level dynamics of nested identities. Rather, they consider the related concept of image and the adjacent forces involved in an entity's sense of self, where the image is represented by the actual perception that others have of an entity and the way in which an entity wishes to be perceived by others (Ashforth et al., 2011).

However, team identity differs from organizational identity, as the value consensus and the socially created rules system enacted within teams are not under the direct control of organizational managers (Larson & Tompkins, 2005). In this regard, our research seeks to provide answers to two questions: (1) What are the cross-level mechanisms between team identity and organizational identity in a modern organization? (2) How can managers and teams' stakeholders influence the formation of team identity?

Research Setting

Agile software development and scrum

Agile software development represents a major deviation from traditional, plan-based approaches to software engineering (Dybå & Dingsøyr, 2008). Its methodologies emphasize the need for lean processes and dynamic adaptation rather than detailed front-end plans and heavy documentation (Nerur & Balijepally 2007). Agile software development is often viewed as the answer to the uncertainties related to detecting, identifying and responding to emerging user requirements (Beck & Andres, 2005; Schwaber & Beedle, 2002). Software development agility rests on a team's ability

to efficiently and effectively respond to changes in user requirements (Lee & Xia, 2010) with minimal formal processes (Cockburn, 2006).

Agile principles, first discussed in the 'Agile Manifesto' (Fowler & Highsmith, 2001), are based on: (1) business-value generation through the delivery of software at regular, short intervals; (2) self-organizing teams whose members work at a sustainable pace, allowing for creativity and productivity; and (3) customer involvement in the development process, which favours feedback and reflection aimed at ensuring more satisfying outcomes (Dingsøyr, Nerur, Balijepally, & Moe, 2012).

'Scrum', which is one of the most common agile methods (Dingsøyr et al., 2012), emerged from the 'new product development game', a holistic approach to product development in which self-managing teams implement innovative products in a relatively short period of time (Takeuchi & Nonaka, 1986). The scrum method prescribes the delegation of decision making to the organization's operational layer and the adoption of self-managing teams. Scrum is suitable in contexts where it is difficult to plan ahead (Dybå & Dingsøyr, 2008) because it involves feedback loops that serve as mechanisms for exercising control over product-release planning processes and for handling changes as a project evolves (Schwaber, 1997). The software is realized in increments ('sprints') that normally last from two to four weeks, each starting with a planning session and ending with a review. The product owner (PO) determines which items are to be developed and listed in the product backlog, which embodies the current knowledge of what needs to be done (Sutherland & Schwaber, 2007). During sprint planning, team members select the top priority items from the product backlog and include them in the next sprint. Daily meetings are held to coordinate the team's work. One team member acts as the 'scrum master' and is responsible for resolving the issues that prevent the team from working effectively.

A telecommunications industry case

Our empirical setting is a large multinational telecommunications firm comprising three interdependent R&D divisions, each relying on agile teams for software production. The software development setting is well suited for a study of team identity in the context of organizational identity construction, as it allows for the simultaneous evaluation of relatively stable between-team factors and more variable within-team beliefs and behaviours. This setting is favourable for an empirical description of team identity construction in the context of a multitude of organization/work-based identifications. As relations and connections can be easily discerned, this setting is also suitable for analysing specific forms of organizational preconditions as well as contradictions in identity construction work.

Team identity construction is constrained by the frequent interactions among team members and by the presence of stable agile routines, which make team boundaries permeable to the environmental context. At the same time, organizational identity construction is affected by the dense interactions between team managers and stakeholders, such as the product owners and the scrum masters. Organizational structures and controls are the same, and similarities in language and culture reduce the likelihood of confounding effects. Organizations developing software products usually work in highly competitive and volatile environments in which time-to-market is crucial. In this context, team identity is continuously affected by ongoing changes in organizational identity and adaptations to environmental instability, which challenge its endurance.

The focal multinational telecommunications firm had a reputation for product and management innovation. The company employed approximately 550 people in the three investigated R&D units; 400 employees used agile software development (mainly scrum). We collected the data between November 2012 and April 2013.

In mid-2011, the company transitioned to agile software development. Before the transition, the three R&D organizations had applied a traditional, plan-driven software development methodology with longer project development cycles and little possibility to dynamically adjust to new customer requests. Extensive upfront planning was used to predict, measure and control problems and deviations for the entire product development process. The approach was process-centric, driven by beliefs about the relevance of highly optimized and repeatable processes related to specific tasks with desired outcomes for each project phase. Roles were assigned to each individual. After transitioning to agile software development, social relations changed, as units were organized around the choices and self-expression of teams. While our aim was not to empirically analyse continuity and change in patterns of social identification throughout the agile transformation, we could identify discrete time periods before and after the transformation.

The transition to agile software development was finalized with the reorganization of human resources into self-managed teams and a ramp-up focused on the agile way of working. In the 18 months prior to our study, no substantial technological changes took place. The three R&D units began the transition to agile software development at the same time and with comparable resources. The agile transformation was launched with the aim of promoting efficiency, learning and cross-functional collaboration within the self-managing teams. Teams devoted to the development of new product functionalities were mostly cross-functional, and made up of designers, testers, software architects and system managers. Cross-functional teams were seen as facilitating product development, as they were expected to reduce information-processing problems, bring a diversity of perspectives to bear on problems, increase problem-solving efficiency and increase the absorptive capacity of the team (Cohen & Levinthal, 1990; Galbraith, 1974).

Research Methods

Data collection

As triangulation from multiple sources at multiple times reinforces confidence in the accuracy of findings (Jick, 1979), we used several data sources: (1) extensive archival data gathered from relevant project and organization publications; (2) 17 group interviews; (3) attendance at multiple team and firm events; (4) repeated semi-structured interviews with executives of the focal R&D units; (5) informal follow-ups through emails, phone calls and observations; and (6) 65 texts generated from open-ended questions in an exploratory survey used as secondary data sources. The survey was launched in the same period as the interviews.

We initially collected qualitative data through four individual pilot interviews conducted with managers in two of the three focal units and through two group pilot interviews with: (1) the head of the firm, the head of innovation at the firm level, and the rest of the management team; and (2) the organizational innovation coaches, who were line managers specifically appointed to promote innovation at the team and organizational levels. The aim of this first stage was to understand the main problems the firm faced after the transformation to agile software development. After agile transition the firm had experienced a year-to-year reduction in the number of patents, system improvements and product ideas. At the same time, even though management did not keep actual data, it had a strong perception that the agile way of working stimulated process improvements, given the high frequency of new process ideas emerging in the organization. We registered examples of innovation behaviours, innovation and learning events in order to evaluate the extent to which the organization was investing in process and product innovation.

After the initial data collection, we interviewed line managers, team members, scrum masters, product owners and system managers in each R&D organization. The interviews lasted from 60 to

90 minutes, and were recorded and reviewed within 24 hours. The interviews covered the following issues: (1) the firm's portfolio formation activities; (2) aspects of the organizational design aimed at achieving operational goals and driving innovation; (3) the strengths and weaknesses of the new organizational design for the implementation of the innovation and operations strategy; (4) organizational routines aimed at enabling knowledge transfer, self-reflection and learning; and (5) perceived barriers to innovation. After the initial interviews, we posed closed-ended questions to a subset of interviewees in order to collect specific factual information (e.g. dates, events, managers involved, issues discussed, deal terms).

We complemented our interview data with observational data. In a three-day visit to two of the three involved units (the first one active in design and the second in testing), we observed six teams working on new telephony software. The third organizational unit supported the design unit and therefore played only an auxiliary role, and it was subject to strict rules and working procedures. Thus, it was only involved in the interview round. We conducted a two-hour workshop with middle and senior managers about perceived barriers to innovation in order to better understand the nature and root cause of the predefined organizational identity. We also analysed the 65 texts derived from a secondary data source (see above). We dealt with potential informant bias in several ways. First, in order to increase the likelihood of capturing a comprehensive, accurate picture we gathered interview data in subsequent waves over five months, and involved informants at multiple hierarchical levels and in the focal R&D units. Second, we relied on informants who were particularly knowledgeable about the organizational design, and the problems faced by employees, and who could reveal part of the organizational culture, identity and logics in order to increase data accuracy (Kumar, Stern, & Anderson, 1993). Third, we guaranteed informant anonymity to encourage candour. Fourth, we triangulated our interview data with archival data. Finally, in order to validate our understanding, we built visual maps tracing all of the claimed problems to the organizational design and organizational characteristics, and discussed them with the relevant organizational stakeholders.

Data analysis

We started our data analysis by focusing on individual case histories derived from interview, observational and archival data (Eisenhardt, 1989). We then triangulated that data. In addition, we checked emails and chat logs in order to gather missing details. Each case was reported in tables that included relevant quotes and organization information. The cross-case analysis was performed by looking at similar emerging concepts and themes (Eisenhardt & Graebner, 2007) and highlighting organizational performance trends that emerged in the years of the agile transformation. This process began after most of the data had been gathered in order to ensure the integrity of the replication logic across the cases (Eisenhardt, 1989). Similarities and differences between cases were identified. The emerging common constructs and themes led to the construction of initial relationships via the replication logic, which relied on continuous efforts to return to each case in order to confirm the occurrence of specific constructs, relationships and logics. This iteration between theory and data eased the refinement of theoretical relationships between emerging constructs (Eisenhardt, 1989). As we identified the theoretical framework, we continuously compared it with the extant literature. We then undertook recurring iterations among data, literature and theory.

Theory and Proposition Development

Main constructs

Team identity. We delineated team identity by adopting attributes that relevant research describes as characterizing identity and that were consistent with our data. Identity can become the

Table 1. Characteristics of team identity.

Data source	Informant	Quotes
Free text in a secondary data source	Team member	'Actually, quality is the top priority inside our team. There is always a need to maintain it, and this has been successfully achieved up to this point.'
Free text in a secondary data source	Team member	'Customer satisfaction and product quality are continuous challenges and exercises for our team.'
Free text in a secondary data source	Team member	'Innovation is viewed as a secondary task that survives in the free space left after our daily work.'
Team group interview	Team members	'I have four projects on my task list that I cannot hand over to somebody else (because there is nobody else who would have time and the right skills), so I cannot work on new ideas.' 'The focus on high efficiency leaves little or no time for innovation.'
Team group interview	Team members	'Working in autonomous teams, but they do not give us any freedom. There is no room left for creativity. We have to be fast. When some ideas pop up, we do not reserve time to realize them. We do not have time to implement ideas.'
Manager group interview	Managers	'People do not have time to carry out innovation. However, beyond the time constraint, it is a matter of establishing the right mindset. We do not know how to motivate people to innovate.'
Archival data	Team	'The retrospective meetings called by team members in order to reflect and improve the way of working were set up with the objective of improving the team's behaviour along the following dimensions: product quality, team spirit, team commitment, tools, agile processes, and relationships among team members.'

'internalized cognitive structure of what the entity stands for and where it intends to go' (Albert et al., 2000 p. 13), especially in the absence of an externalized bureaucratic structure serving as a 'rudder for navigating difficult waters' (Albert et al., 2000, p. 13). In most of our focal cases, the identity of autonomous teams was built on the need to satisfy project tasks and accommodate proximal managers and stakeholders. Table 1 summarizes the qualitative characteristics of the identities of the sampled teams.

Organizational identity. Organizational identity was detected through archival data, and through its main contributors' (i.e. top managers) descriptions of the current and intended direction for the organization. Related evidence showed close alignment between the established team identity and the organizational identity shared among top managers. Identity was focused on the need to ensure customer satisfaction and, to some extent, on product innovation. Table 2 summarizes the qualitative characteristics of the organizational identity.

Team stakeholders. By team stakeholders, we mean the individuals who have an interest in leading a team towards the desired performance. Table 3 lists the identified teams' stakeholders together with their relevant roles in the teams' performance, based on the available data and the observations reported.

Figure 1 represents the organization design built up around the teams, including the identified stakeholders, and their connections with each team and their peers. Figure 2 presents the organizational form selected to drive product innovation inside each unit. It assumes that only one person

Table 2. Characteristics of organizational identity.

Source of data	Informant	Illustrations
Archival data: Operational plan for the entire organization	High-level management	The yearly organizational operational plan did not emphasize product/technology innovation and learning, but only the dimensions of motivation, recognition among peers, agile, increased collaboration, software (SW) craftsmanship, customer engagement, the quality concept, and competence. Efforts to increase competence were planned with the aim of filling gaps in the following areas: continuous SW integration, SW craftsmanship, SW development expertise, people leadership, team culture, consulting skills, and customer orientation.
Archival data: Performance management philosophy in the organization	High-level management	Performance management measures were conceived at three levels: individual, team and organization. At the team level, they included budget and delivery precision for new feature-development activities as relevant metrics. Quality checks were planned at the end of the feature development and of the sprint via the 'definition of done' documents. Key performance indicators for each product area were used to monitor the feature development at higher levels in the organization. Organizational motivation was assessed via an annual survey and constantly checked with people in the team through regular one-to-one talks. There were no formal indicators to monitor teams' ideation capabilities, or the time spent on learning or creative activities. Product-innovation performance was measured at the unit level and across units at the site level. It was measured as the number of patents.

Table 3. Teams' stakeholders.

Source of data	Stakeholder	Illustrations
Archival data: Description of the role found in the organizational documentation	Scrum master	• 'Drive the evolution of the team towards high performance • Ensure that all team members are integrated and contributing • Provide feedback on team performance and individual development needs • Drive the implementation of agile methods • Ensure that the scrum framework, and lean and agile principles are applied • Actively remove impediments'
Archival data: Description of the role found in the organizational documentation	Product owner	• 'Provide the product vision and translate and transfer requirements towards the team • Own, maintain, and prioritize the User Stories Backlog for the team • Define the 'definition of done' for the feature responsible'
Archival data: Description of the role found in the organizational documentation	Line responsible	• 'Act as a lean/agile agent for the team • Secure right team composition with a focus on competence • Secure the work environment for the team • Provide important feature information to the team • Maintain contact with team members (and product owner and scrum master) to feel the 'temperature' within the team and act on it • Support the scrum master in issues regarding team performance, team efficiency, team development and team setup • Serve as the first escalation point for the scrum master in terms of handling impediments • Ensure that team members build cross-functional competencies'

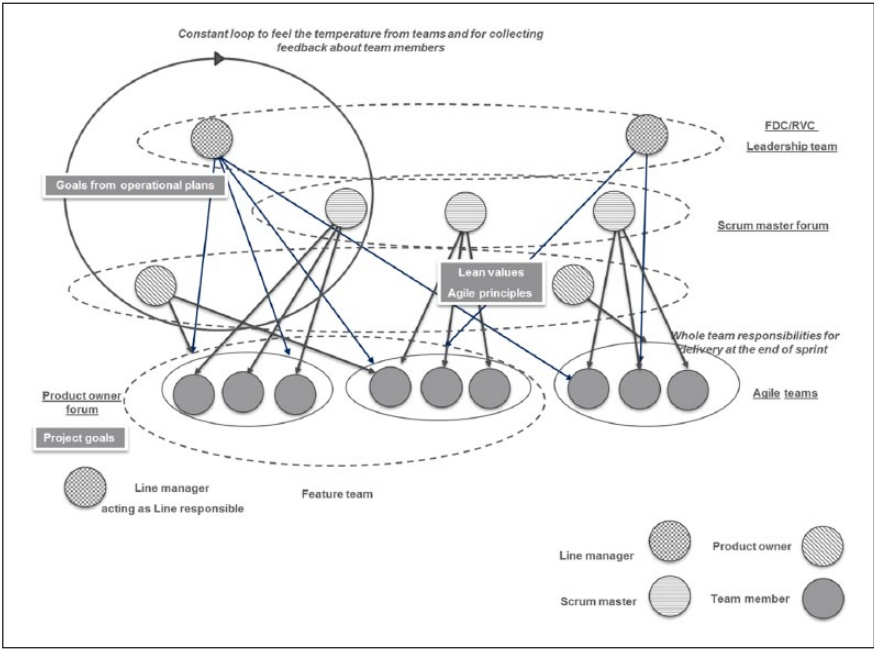


Figure 1. The stakeholder network of teams.

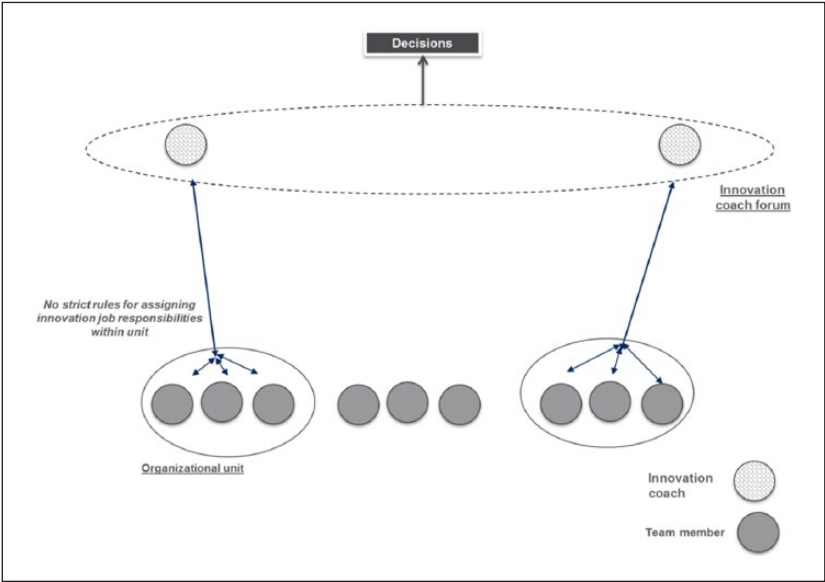


Figure 2. Organization design for driving innovation.

in each R&D unit organization takes on the role of innovation coach. These coaches support innovation ideas coming from the teams, provide them with infrastructure and inspiration for innovation, create learning opportunities and establish workshops with customers, and offer logistics

support, which is consistent with the organization's innovation strategy. This is a secondary role for a line manager that demands only a small percentage of working time. These coaches report directly to the unit's management team. Both figures are based on our field observations, inspections of organizational documents, and interviews, and they have been validated by key organizational informants.

Forming and strengthening team identity by orchestrating controls and working routines

Relational networks are crucial for the creation of institutional values and logics in an organization (see Owen-Smith & Powell, 2008). Intense interconnections among individuals in an organizational network enable efficiency in communication and favour the exchange of information across the network in a way that allows norms and expectations to spread and become institutionalized (Meyer & Rowan, 1977). An increase in density in the stakeholder network (e.g. the number of interactions in a network linking actors together relative to the total number of possible interactions) leads to shared behaviours and expectations among members, resulting in the organization's actions being affected by normative mechanisms (Rowley, 1997). Dense networks create strong, unified stakeholder pressures, which drive the organization towards conformity.

Our data indicated that the routines characterizing agile software development made the teams' boundaries permeable to the external environment by allowing team stakeholders to frequently and regularly interact with the teams. Teams' stakeholders were present, albeit silent, during daily scrum meetings at which team members reported on their progress. Scrum masters, who were responsible for removing any impediments to the team's work, were key actors in these meetings.

At the end of each sprint, teams showed their accomplishments to the product owner and the line manager during a formalized sprint review. The product owner sought to verify whether the produced deliverables satisfied the criteria established during sprint planning. Group performance was constantly monitored by the scrum master with the aim of triggering reflection on ways to improve teamwork. The product owner and the scrum master provided continuous updates to the line manager. The line manager regularly interacted with the scrum master, who was the first escalation point if impediments were severely affecting team performance. These interactions provided constant information on team performance and operative needs.

Teams' stakeholders were tasked with spreading agile practices and values, and with driving the transformation to agile software development. The dense networks of stakeholders had the capacity to efficiently monitor the team's actions, thereby applying a form of pressure on the team and pushing it towards conformity. This was underpinned by the significant overlap between team and stakeholder identities. In contrast, in the less dense network in which innovation coaches drove product innovation, teams had more discretion over their actions, as they were under less unified pressure. While the sparse, fragmented network of the innovation coaches exhibited a limited ability to monitor team behaviours or to create shared behavioural norms, project stakeholders were able to exert a strong influence on teams and the resulting pressure sometimes conflicted with that of the innovation coaches. Consequently, teams were often unable to conform to innovation coaches' expectations because doing so would require defying the expectations of project stakeholders. As such, teams could resist innovation coaches by claiming a lack of time for innovation and by arguing that the quality of the product was the priority. The above reasoning and observations motivate two propositions:

Proposition 1: *In the presence of agile routines, the combination of shared expectations that agile practices will be followed and the ease of information exchange among an agile team's stakeholders create conditions for applying strong, unified stakeholder pressure to teams.*

Proposition 2: *The denser a team's stakeholder network, the greater the influence of that team's stakeholders' needs, values and beliefs on the team's reconstruction of organizational identity, so that the team's identity might become isomorphic with organizational identity.*

A focal actor's centrality can influence that actor's degree of influence over stakeholders (Rowley, 1997) due to the actor's prominence (Wasserman & Galaskiewicz, 1994) or power (Brass & Burkhardt, 1993). Three types of centrality, referring to different aspects of actors' positions, are usually considered in the social network literature: degree, closeness and betweenness centrality (Brass & Burkhardt, 1993).¹

According to the archival data, the scrum master had to provide 'feedback on team performance and individual development needs' to, respectively, a line manager responsible for the focal team, and a line manager responsible for team members' growth and career development. In addition, the scrum master was in contact with the person responsible for the new software's functionality (i.e. the 'feature') and the product owner through regular meetings designed to inform the person responsible for the feature about the 'temperature within the team'. In addition, the scrum master had to escalate all of the issues and impediments raised by the team to the line manager, who was expected to 'support the scrum master in issues regarding team performance, team efficiency, team development, and team setup.' Figure 3 shows that the scrum master holds a central position and has access to all other actors in the network. The line manager is two steps away from the team, while other actors are one step away.

As the scrum master had the shortest aggregate distance to the other actors, he or she could quickly diffuse information. Centrality is related to effective communication, as closeness 'mean[s] fewer message transmissions, shorter times and lower costs' (Freeman, 1979 p. 225). With regard to betweenness centrality (the degree to which an actor exercises control over other actors' access to different areas of the network; Freeman, 1979) Figure 3 suggests that the scrum master was again highly central, as all actors had to pass through him or her to exchange information with other parts of the network.

An actor in a central position is able to influence behavioural expectations and mediate information flows within the network, potentially biasing the communication to the actor's advantage. The scrum master was favourably positioned to isolate stakeholders from one another and to undermine their coordination abilities. Consequently, his or her imposed norms and beliefs were diffused across the stakeholders' networks. The archival data suggest that scrum masters should: (1) 'educate and train the organization in agile and lean principles'; (2) 'learn quickly from the features delivered'; (3) 'support feature teams in their efforts to be more productive in any way they can'; (4) 'secure active and efficient communication among feature teams'; (5) 'encourage transparency and sharing of good practices'; (6) 'foster a feedback culture'; and (7) 'address and report on organizational and cross-feature impediments'. The group interviews confirmed that scrum masters' attention and focus reflected their desire to make teamwork smoother, favouring improvements and, eventually, learning in order to implement new functionalities within their development projects. An organizational document explained the vision for the scrum masters:

Scrum masters are a community that fully understands the mechanisms and the processes that drive our product forward in an agile environment [...]. We assist the teams in challenging the status quo and pursuing simplicity, so the organization is able to understand and learn the processes that can drive us in developing products leveraging on motivated individuals.

In crafting the organization's operational plan (i.e. the implementation of its business-level strategy), managers highlighted the following dimensions of interest: agile processes, agile values related to the idea of process improvement, product quality (as prescribed by lean principles), team

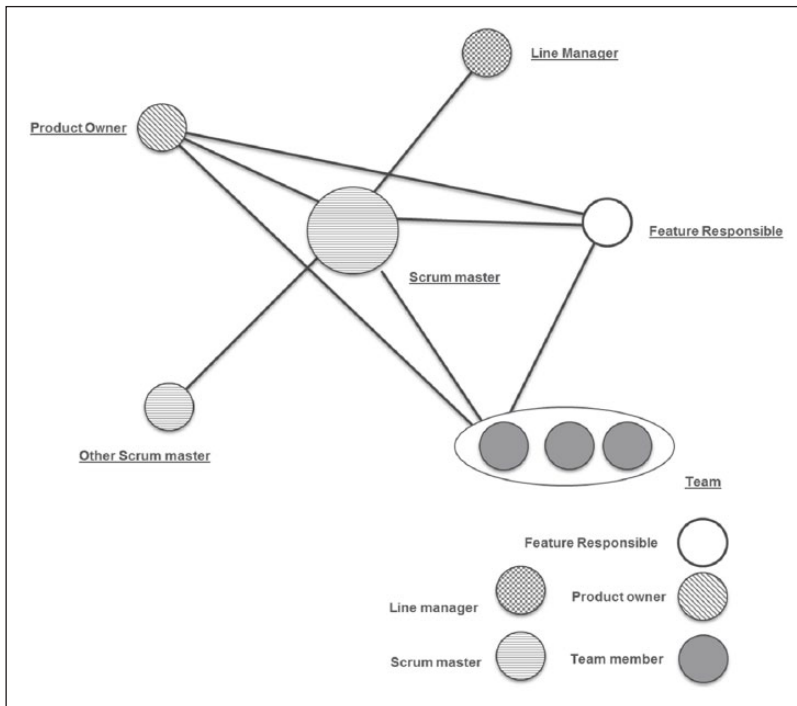


Figure 3. Scrum master's position within the stakeholders' network.

collaboration, team members' technical excellence, and customer involvement (as prescribed by agile). The project performance system was centred on several dimensions: cost, speed, product quality and customer value; the adoption of budget checkpoints on specific feature 'toll gates'; speed checkpoints using burn-down charts; and value assessment checklists and quality checks at the feature and sprint levels. The latter were implemented using 'definition of done' documents, a list of actions to complete before a focal piece of software would be ready to be released. Team outcomes were checked externally for product quality, project delivery times and competences, and each team's internal performance was continuously monitored and supported by the scrum masters. This was confirmed by minutes from retrospective meetings in which some team members stated: 'The scrum masters press the team to produce and to deliver.' In other words, scrum masters imposed and diffused norms and beliefs about the need to improve processes and to act according to agile practices across the stakeholders' networks. These findings suggest the following proposition:

Proposition 3: *The greater the centrality of a focal actor, such as a scrum master, in the team's stakeholder network, the lower the influence of other stakeholders' needs, values and beliefs on organizational identity.*

As suggested in Figure 4, the scrum master played a special role inside the team due to his or her regular and frequent interactions with team members. As shown in Table 3, the product owner had a smaller impact on team members' actions than the scrum master, although that impact remains relevant.

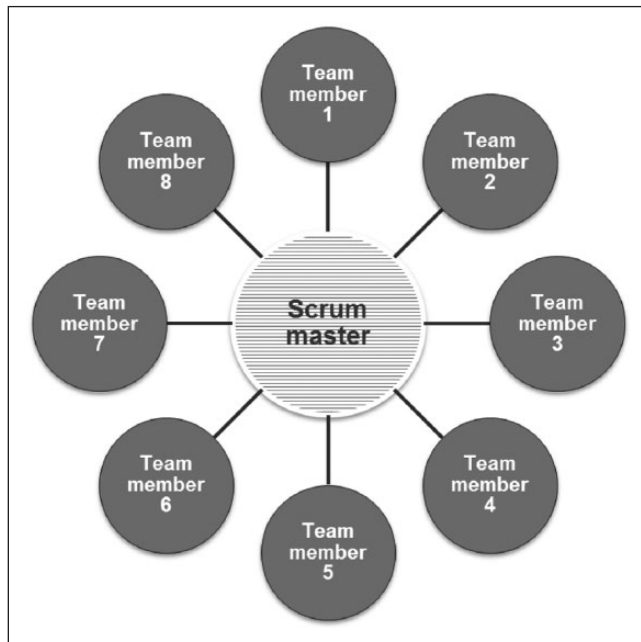


Figure 4. Scrum master's position within team.

An increase in interactions with the organization through its stakeholder groups raises the strength of organizational identification (Dutton et al., 1994). Organizational identity mirrors a negotiated order (Strauss, 1978) that take shape as members work together, communicate their preferences, show their problem-solving styles (Swidler, 1986) and manage external demands and internal needs for coordination and integration (Schein, 1990). This shared experience can generate a common and rooted (Denison, 1996) understanding of how the social environment works, and determine which practices and standards are defined as appropriate and effective. Identity represents an aspect of the organizational environment that acts as a reference, and helps members make sense of their own and others' behaviours.

Greater interaction among stakeholders and teams increases the accessibility and salience of stakeholders' organizational social identity, which is likely to influence how team members evaluate and respond to stakeholder issues. This occurs, first, by creating a common interpretative frame for the collection, screening and evaluation of information about stakeholder attributes and, second, by motivating behaviours and practices that protect, improve or support the organization's identity. The central role played by stakeholders such as the scrum master helped resolve identity ambiguity and allowed team members to make sense of the new identity (Corley & Gioia, 2004), perhaps by brokering information (Rowley, 1997). Team members might rely on those stakeholders with higher perceived power, legitimacy and urgency (Mitchell et al., 1997).

In the organizational context, scrum masters were given power and legitimacy to operate from the focal R&D organizations. Their responsibilities were described in organizational documentation:

- (1) drive the evolution of the team towards high performance, (2) drive the implementation of agile software development, (3) provide feedback on team performance and individual development needs, (4) ensure application of the scrum framework, and (5) remove specific impediments.

When discussing the role of the scrum master and the centrality of his or her position relative to the team, one team member reported:

The scrum master triggers reflections on how to improve the speed of flow and new ideas of way of working. In particular, the scrum master stimulates us during the daily stand-up meeting, giving us suggestions for improving the way we organize the work for our deliveries.

Scrum masters gained legitimacy when team members view their actions or claims as proper and appropriate given the prevailing standards of the institutional context. Notes from teams' retrospective meetings show that constant attention was paid to the teams' adherence to agile software development in line with the scrum master's focus. We therefore propose:

Proposition 4: *The embeddedness of a scrum master within a team favours the diffusion of the scrum master's norms and beliefs inside the team. Therefore, team identity and alignment with the organization identity are moderated by the presence of a scrum master.*

Mechanisms for team identity regulation

Team self-regulation is a self-monitoring process aimed at changing the team's cognition and affection. It may include shaping the organizational identity, especially if this is needed to strengthen the team's efficiency. Behavioural self-regulation refers to self-observation and strategic adjustments to expected goals, actions and reactions. This perspective embodies the team cognition and its interpretation of organizational identity, which enhance or constrain engagement and participation in realizing the stakeholders' objectives.

Team and social regulatory processes can be diverse and located at different organizational levels, but they can function together to influence a team's perception of organizational identity and development through 'co-regulation' (Fogel, 1993). This macro-level perspective offers a broader picture in which co-regulation (team control and social influences) affects a team's cognition and affection through opportunities to operate and receive feedback within a social context (McCaslin, 2009).

In our study, the adoption of the scrum methods led to the implementation of a broad system of interrelated feedback controls and the adoption of numerous routines for teamwork. For example, the product owner prescribed the team's tasks for a specific timeframe, then checked and weighed that work against specific criteria at the end of each two-week sprint. Teams were required to continuously report to the product owner on the status of their activities. At each retrospective meeting, the scrum master measured and assessed team performance in terms of speed in delivering user stories on the basis of burn-down charts. Working in isolation from the rest of the organization, agile teams were responsible for releasing new software functionalities every two weeks, which were then integrated into the overall product. Automatic regression tests were launched almost daily to check for the integrity of the product system, while team members worked towards having a demonstration ready for their stakeholders at the end of each sprint. Quality checks were undertaken at the end of each feature and/or sprint. 'Definition of done' documents were handled by the product owner and by those responsible for the team and the feature. The team's budget and delivery precision were constantly measured and reported to higher levels, and team competences were regularly assessed in order to detect gaps.

This example illustrates how control of team members' behaviour may be realized: sanction mechanisms aimed at limiting deviating behaviours were routinely part of the jobs of organizational members, teams' stakeholders and managers. Team self-regulation does not occur in a

vacuum: many parties have an interest in a team's actions and decisions. Self-regulation, which is grounded in a control theory framework (Carver & Scheier, 1981), suggests that teams adopt standards that reflect their perceptions of the organizational image, test their cognitive understanding against those standards using information taken from their social environment, and take actions to decrease any discrepancies. The team's regulatory aim is to reduce discrepancies between standards and 'enacted' behaviours (Lord & Hanges, 1987).

Information and feedback from stakeholders enable the team to reduce discrepancies and adjust behaviours, thereby renewing the interpretation of the organization image. When stakeholders' behaviours are supportive of team goals and actions, they reinforce team perceptions of organizational identity in a direction that is consistent with desired organizational images and the team becomes more resilient to change (Bem, 1972). A team member stated, on this point: 'The problem is that there are too many stakeholders and different perceptions of what we are doing, while no one is justifying the time we may spend on learning or the things we are learning.' Team members added that their main priorities were the features on which they were working. Teams' beliefs seemed aligned with the scrum masters' mission 'to learn fast from the features delivered,' which was reinforced by the organizational expectations for the scrum master's role: 'drive the evolution of the team towards high performance' and 'ensure that all team members are integrated in and contributing to the scrum framework, and that lean and agile principles are applied'.

Stakeholders' continuous validation ensures their pivotal role in the building of team memory. In light of convergent information from stakeholders, team identities, which are formed in particular situations with certain stakeholders, eventually become generalized to other situations and stabilized, so that they are cued in team members' memories. This suggests the following proposition:

Proposition 5: *Team members engage in active processing of an organization's identity-relevant information to reflect stakeholders' appraisals, managers' self-identities and perceived organizational goal attainment.*

Though quite strict, the feedback-based control systems can leave members with areas of behavioural discretion, creating an opportunity to be out of sync with the current organizational perspective. An example of this was given by a team member:

During the grooming session, our product owner asks us to only consider the possible investigations needed to implement features. Even then, the time needed for this type of investigation/competence building differs from sprint to sprint but, on average, it is probably around 10% of the sprint time. Actually, there is no real time allowed for our competence building-up from our stakeholders. Training on certain tools and methods is sometimes arranged.

The acceptance of the organizational identity can be fostered in order to reduce the extent of team members' discretion and decision making. This limits the range of options to those congruous with that identity (Tompkins & Cheney, 1985) and organizational goals. Discretion can be restricted by the expectation that behaviour should mirror organizational values (Sewell, 1998).

Interview and archival data suggested that daily stand-up meetings run by team members in the presence of line managers and stakeholders served as a rule enforcement mechanism. These meetings made routine operations possible and hindered team members from pursuing their own interests, which could be antithetical to the goals of the organization and the team. Team members attended daily scrum meetings together with the relevant stakeholders and managers, who were only allowed to listen. Each team member had to answer the following questions: *What did you do*

yesterday? What will you do today? Are there any impediments? By focusing on each person's accomplishments, the team gained an understanding of what was completed and what remained to be done. Given the high level of interdependency among members and the presence of a scrum master fostering knowledge of agile routines, team members could reach a common perspective by relying on shared task knowledge and common flow expectations. The silent presence of managers and stakeholders was recognized by the team in the decision-making process, while the organization made a subtle statement to individuals of its values regarding team autonomy. These actors used meaning-laden actions to inform team members of their values and to raise the attractiveness of organizational images among team members, thereby encouraging a favourable self-image (Dommermuth, 1989). Our field observations suggested that team members engaged in bricolage (Bechky & Okhuysen, 2011). Team members publicly drafted agreements on their work with the participation of managers and stakeholders, and created opportunities for team members to ask questions, propose alternatives or critique approaches. These 15 minutes of active and engaged conversation took place every day in view of everyone. The temporal flow of the work, which imposed a constant rhythm with frequent deliveries, the continuity of team memberships, the consequences for the team and the organization if deadlines were not met, and the constant presence of stakeholders with similar interests in team performance and behaviours framed the ways in which team members responded to urgencies and built the necessary resources.

Managers and stakeholders influenced the team's identification with the organizational identity through two independent routes: (1) showing a desired organizational image that was attractive to team members during the daily stand-up meeting (e.g. highlighting opportunities that the organization offered to individuals to express themselves; see Shamir, 1991); and (2) influencing the extent to which team members' participation in the team attracted their attention. While some organizations advocated worthy social issues to enhance their legitimacy, managers and stakeholders strived to boost legitimacy by incorporating qualities they believed as particularly valued by team members and by deflecting attention from negative aspects of their identity. The presence of managers and stakeholders also gave legitimacy to teams' agile routines, leaving team members with the perception of being compliant with organizational goals. Legitimacy was held by the team 'independent of particular observers' (Suchman, 1995, p. 574). Nevertheless, team members' attributions of legitimacy mirrored their individual beliefs that relevant others recognized a congruence between the team's behaviour and '[their] shared (or assumedly shared) beliefs' (Suchman, 1995, p. 574). The public presentation of the team's activities, and planning in the presence of managers and stakeholders, constituted important mechanisms for improving individual identification with the organization.

The visibility of the affiliation with the organization moderates the relationship between the attractiveness of a perceived organizational identity and organization identification, as visibility supports people's affiliation within the organization (Dutton et al., 1994). On the other hand, the visibility of team's participation in the realization of its stakeholders' objectives generates an enhanced cognitive elaboration of the organizational social identity owned by the stakeholders. Based on this reasoning, we offer the following proposition:

Proposition 6: *The visibility of a team's work processes favours the cognitive elaboration of a perceived organizational identity within the team, thereby increasing the accessibility and salience of team-related stakeholders' organizational social identity and the likelihood that team members will eventually become self-disciplined.*

Team members are involved in self-construction processes within their team through which they try to define who and what they are. The high interdependence within the team intensifies the

connectivity among people. As connectivity becomes more intense, team members tend to behave similarly despite their freedom to be different. This creates a set of legitimized working conventions (Scott, 1995) nested in the value-based controls that becomes the basis on which members operate in the development of products or services. Shared task knowledge is developed, making it easier for team members to coordinate without explicitly communicating (Cannon-Bowers & Salas, 2001).

The longer one stays within an organization, the more prominent self-categorization and organizational membership become (Kramer, 1991). As identification with a group increases, so does the motivation to reach group goals (Ashforth & Mael, 1989; Brewer & Gardner, 1996; Dutton et al., 1994). When members socially identify with their team, they emphasize similarities among in-group members and try to demonstrate differences with non-group members (Hogg & Abrams, 1988). Moreover, once socially identified, members evaluate the team's claims as legitimate (Brown, 1997). In that case, team identity can enhance individual self-esteem, leading to a perception of group identity as legitimate and as 'desirable, proper, and appropriate within some socially constructed system of norms, values, beliefs, and definitions' (Suchman, 1995, p. 574). The following proposition sums up this reasoning:

Proposition 7: *Team members' task interdependencies and team members' stability inside the team favour individual identification with the team identity.*

Within the team, individuals are exposed to the scrutiny of the other members, so that low performers are motivated to improve their efforts and high performers push others towards continuous improvement. In the daily meetings, team members had the opportunity to indicate their approval or disapproval of the behaviour or performance of others, making the daily meeting a forum for rewards and sanctions. This points to a controlling, horizontal, disciplinary force running across the team, creating the conditions for an extended mode of concertive control (Barker, 1993). Concertive control is based on the ability of the team to use its own sanctions and rewards, and is associated with the individual basis of evaluation discussed above (Sewell, 1998). The concerted actions of groups do not affect team conformity to norms (i.e. accepting team identity as one's own) (Hopkins, 1964). Rather, they can affect compliance with those norms (i.e. behaving in accordance with the current team identity). Concertive control is directly correlated with compliance (Barker, 1993).

In our context, concertive control was enhanced through the scrum master's surveillance. As the scrum master was embedded in the team, he or she had an organizational mandate to reach a high degree of compliance, implementing what Sewell (1998) calls 'biopower'.² Team members indicated that they suffered from high levels of pressure after the introduction of agile software development. In these conditions, environmental features more central to completing the task seemed to increase in salience and vice versa, leading team members to interpret the goals of their interactions differently and to use different criteria to judge the quality of the group product. Groups working under time pressure worked with certain modes of influence, such as 'normative' or 'informational' influence, suggesting a focus on task completion and consensus (Kelly, Jackson, & Hutson-Comeaux, 1997). Normative influence favoured consensus formation when resolving differences of opinion, while informational influence favoured consensus formation when resolving differences of fact. Our data revealed the presence of concertive control on teams, which is a normative influence. As normative influence is associated with the heuristic processing of information (Kaplan, 1987), time pressure reduced the systematic processing of task information, leading to faster decisions.

Time pressure might have led team members to focus on a narrow range of task-relevant cues and to embrace task completion as the relevant interaction objective. As a team member observed:

'It is impossible to have a clear mind for innovations when a new product is developed. You are always under pressure to succeed with the next delivery. Therefore, there is very little time to think about new ideas.' As a result, team members mostly talked about task-relevant information in their interactions and processed such information less systematically, as indicated by the following statement by a team member:

The greatest obstacle to innovation is that we are overloaded with tasks in our day-to-day jobs and we do not get a chance to work on innovative areas. We are just given meaningless goals to produce meaningless ideas, which are measured in numbers and not evaluated in terms of their value for the company.

Group members were worried about achieving consensus or making a decision quickly, and not concerned with other goals, such as improving innovation performance or systematically evaluating alternatives. One team member highlighted the lack of routines addressing decisions on innovation: 'No space and time for it. Not an integrated part of daily work. Not enough recognized and encouraged examples. We have a graveyard of many good ideas without knowing what happens with the selected ones.' These interactions and information-processing strategies led to less creative, less satisfactory decisions.

The above reasoning and observations suggest the following propositions:

Proposition 8: *The team's concertive control and the scrum master's presence in the team jointly cause continuous re-establishment of team identity towards higher performance standards and full compliance of the team members with the team's renewed norms.*

Proposition 9: *Team-internal tensions with regard to enhanced performance standards and full compliance with the team's norms create a feeling of pressure inside the team and focuses the attention of team members on the organizational goals.*

Concluding Discussion

In this study, we have analysed mechanisms of team identity given the socially constructed nature of organizational identities. This joint construction provides an understanding of the dynamic relationships among organizational entities, and legitimizes their influence on the mechanisms of organization and team identity formation. Based on data from small-N research in a single firm and on extant theory, we developed theory on how the dynamics of nested organizational identities evolve across organizational levels, focusing on mechanisms facilitating identity cascades. In an organization adopting agile software development, nested identities gradually become more focused on and shaped by the established organizational identity as one descends the cascade.

At a given level below the organizational level, different organizational entities emphasize diverse attributes of the higher-order identity. In the focal organization, the structural differentiation principle was applied: entities on the same level focused on certain aspects of the inclusive higher-order identity. For example, scrum masters focused on the team's adherence to agile practices with little tolerance for deviations from the plan, while line managers focused on motivational aspects of team members and on the competences needed for project development activities.

Although clear group identities directly informed and legitimized relational identities among individuals within the group, there could be some variance across relationships, as relational identities imply individual differences (Sluss & Ashforth, 2007). Research shows that individual-level characteristics that help differentiate group members are pivotal within the context of the group's identity (Postmes & Jetten 2006). Entities at the lowest level of analysis may intentionally diverge from their prescribed identity (Foreman & Parent 2008) for purposes of self-aggrandizement

(negative deviance). Agile routines (e.g. daily stand-up meetings, regular feedback loops) force compliance with the team's norms and ensure convergence towards the organizational identity. The goal is to reduce the likelihood that entities might develop strategic ambiguity and accommodate diverse preferences. Discursive resources, derived from the constant presence of the scrum master, are used to promote, ratify and reinforce the convergence of identity claims inside the team. They also reinforce isomorphism across organizational levels (Sveningsson & Alvesson, 2003).

While most research on organization-based identities concentrates on a single level of analysis (the individual, the group or the organization), we addressed the question of cross-level mechanisms between team and organizational identity construction, and offered a speculative discussion concerning nested identities. In this regard, this paper advances our understanding of the processes through which identities are linked across levels, and explores how identities at one level of analysis can favour and constrain identities at other levels.

Building on these linked identities, we identified ways in which they become isomorphic across levels and ways of overcoming various barriers to isomorphism. We highlighted the importance of understanding the interrelationship between the projection of team identity and the feedback received by different organizational members. In addition, we viewed organizations as fractured and hierarchical locales in which individuals and groups are reciprocally involved, but often through asymmetrical power relationships (Clegg, 1981). Consequently, the exercise of power affects how organization and team identities develop. In order for these identity construction processes to be examined, we needed to analyse the influence and interplay of the identities that characterize individuals, teams and organizations (Brown, 1997). This approach allowed us to provide an interpretation of patterns of interaction belonging to a specific organizational design. Investigations of the degree of resistance to change at both the team and organizational levels would be an interesting avenue for research.

Our arguments are relevant for understanding organizational change in post-bureaucratic structures relying on self-management but coordinated by a higher organizational layer. It seems reasonable to suggest that the degree of a team's resistance to change is an integral aspect of its identity and that strategic change implies identity change (Gioia & Thomas, 1996).

The difficulties of regulating team identity documented in this paper show that understanding the inertial power of post-bureaucratic structures might inform research in this field. Familiarity with processes of team identity change may improve our understanding of how organizations reshape and reorder their strategic discourse. We have analysed the power of identities as a form of behavioural control, thereby enhancing the classic typology of control. We did so by clarifying the link between desired organizational identities and a form of control that leads to team members' moral obedience. Individuals with collective identities identify with and internalize their group's goals and norms, and are therefore committed to accomplishing their responsibilities as group members (Johnson & Yang, 2010). Research has shown that organizations can influence the formation of organizational members' identities (Alvesson & Willmott, 2002; Van Maanen & Schein, 1979), but it typically fails to explain exactly how this occurs. The possibility that identities may be triggers for action has been mentioned but not elicited or detailed, with the exception of a few notable contributions (White, 1992; Akerlof & Kranton, 2005).

Our study suffers from some limitations and highlights avenues for further research. First, we point to the issue of 'moral obedience' (Courpasson & Dany, 2003), which is important in a context of empowerment, as deserving of more research attention. We focused on the self-managed, autonomous team as a relevant unit of analysis, but it would be interesting to observe individuals in teams who may be morally obliged to conform with decisions over which they have no real control. Extant research (Courpasson & Dany, 2003; Heckscher & Donnellon, 1994) has analysed the control of empowered people working and acting within consensually based systems of power. The

evidence seems to suggest that the dilemma related to obedience depends on how people view disagreement and disobedience given the strict relations among bureaucracy, community sense and social dynamics. A position maintained on moral grounds is related to identity status. Thus, team identification could be a means to realize individual moral obedience (Podd, 1972).

Second, we analysed how a team relates to diverse social categories when constructing a sense of self (see Alvesson & Willmott, 2002; Sveningsson & Alvesson, 2003). The intersectionality perspective suggests that identity construction takes place in the relations between different organizational categories (Essers & Benschop, 2007). We have focused on inequalities in organizational and team identity construction, and analysed how organizational stakeholders and teams relate themselves to the privileges and/or disadvantages of prominent categories (Zanoni & Janssens, 2007). However, we analysed silence as a discursive phenomenon related to excluding potentially important understandings on a specific topic and building knowledge on it. Here, silence can be seen as related to different ways of neglecting issues related to social categories. On the interpersonal level, social categories are seen as related to others, meaning that people are mostly silent on themselves as members of categories. On the individual level, silence is related to what is said and not said about oneself as a member of social categories. In both cases, there is a tendency to avoid inequalities: at the organizational level, by not referring to inequalities at work; at the individual level, as disadvantages and/or privileges and their impact on oneself at work.

Third, the analysis of context is in need of further development. The same is true for the means through which elements of identification may be changed or manifested when both proximal and distal circumstances are different from scrum implementation.

Finally, we have explicitly acknowledged the roles of organizational practices, norms and stakeholders' and managers' interactions in giving substance to collective self-perceptions in teams, and we have highlighted how these manifestations affect team identity formation. Nonetheless, how current organizational culture (i.e. the traditional way of working) influences responses to identity threats remains unexplored. Future research may deepen our understanding of the evolution of organizational identities through environmental changes. Along these lines, we suggest that collective history, organizational symbols and consolidated practices may complement our analysis.

However, as this study provided a wide context for an examination of identity formation across multiple levels, and for the analysis of organizational routines and roles, we believe that the approach used herein might enable scholars to better understand these phenomena. We hope that the use of our conceptualizations will help to effectively trace identity dynamics, such as how collective identities emerge and change over time.

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Notes

1. Degree centrality can be described by the number of ties an actor has with other actors in the network. Closeness describes an actor's ability to independently gain access to all other actors in the network. Betweenness measures when an actor is in the geodesic paths between pairs of other actors (Freeman, 1979).
2. 'Biopower normalizes individuals within categories through its ability to identify prodigious subjects, people who don't fit in. These misfits would then receive the attention of the normalizing forces of modern society, such as therapy, discipline, or medical treatment' (Sewell, 1998, p. 421).

References

- Albert, S., Ashforth, B. E., & Dutton, J. E. (2000). Organizational identity and identification: Charting new waters and building new bridges. *Academy of Management Review*, 25, 13–17.
- Albert, S., & Whetten, D. A. (1985). Organizational identity. In L. L. Cummings & B. M. Staw (Eds.), *Research in organizational behavior* (pp. 263–295). Greenwich: JAI Press.
- Akerlof, G. A., & Kranton, R. E. (2005). Identity and the economics of organizations. *Journal of Economic Perspectives*, 19, 9–32.
- Alvesson, M., & Kärreman, D. (2001). Odd couple: Making sense of the curious concept of knowledge management. *Journal of Management Studies*, 38, 995–1018.
- Alvesson, M., & Willmott, H. (2002). Identity regulation as organizational control: Producing the appropriate individual. *Journal of Management Studies*, 39, 619–644.
- Ashforth, B. E., & Mael, F. (1989). Social identity theory and the organization. *Academy of Management Review*, 14, 20–39.
- Ashforth, B. E., Rogers, K. M., & Corley, K. G. (2011). Identity in organizations: Exploring cross-level dynamics. *Organization Science*, 22, 1144–1156.
- Austin, R. D., & Devin, L. (2009). Research Commentary. Weighing the benefits and costs of flexibility in making software: Toward a contingency theory of the determinants of development process design. *Information Systems Research*, 20, 462–477.
- Barker, J. (1993). Tightening the iron cage: Concertive control in self-managing teams. *Administrative Science Quarterly*, 38, 408–437.
- Bechky, B. A. (2011). Making organizational theory work: Institutions, occupations, and negotiated orders. *Organization Science*, 22, 1157–1167.
- Bechky, B. A., & Okhuysen, G. A. (2011). Expecting the unexpected? How SWAT officers and film crews handle surprises. *Academy of Management Journal*, 54, 239–261.
- Beck, K., & Andres, C. (2005). *Extreme programming explained: Embrace change*. Boston, MD: Addison-Wesley.
- Bem, D. J. (1972). Self-perception theory. In L. Berkowitz (Ed.), *Advances in experimental social psychology* (pp. 1–62). New York: Academic Press.
- Brass, D. J., & Burkhardt, M. E. (1993). Potential power and power use: An investigation of structure and behavior. *Academy of Management Journal*, 36, 441–470.
- Brewer, M. B., & Gardner, W. (1996). Who is this ‘We’? Levels of collective identity and self-representations. *Journal of Personality and Social Psychology*, 71, 83–93.
- Brown, A. D. (1997). Narcissism, identity, and legitimacy. *Academy of Management Review*, 22, 643–686.
- Cannon-Bowers, J. A., & Salas, E. (2001). Reflections on shared cognition. *Journal of Organizational Behavior*, 22, 195–202.
- Carver, C. S., & Scheier, M. F. (1981). Self-consciousness and reactance. *Journal of Research in Personality*, 15, 16–29.
- Cheney, G., & Christensen, L. T. (2000). Organizational identity: Linkages between internal and external communication. In F. Jablin & L. Putnam (Eds.), *The new handbook of organizational communication* (pp. 231–270). Thousand Oaks, CA: SAGE Publications.
- Clarkson, M. E. (1995). A stakeholder framework for analyzing and evaluating corporate social performance. *Academy of Management Review*, 20, 92–117.
- Clegg, S. (1981). Organization and control. *Administrative Science Quarterly*, 26, 545–562.
- Cockburn, A. (2006). *Agile software development: The cooperative game*. New York: Pearson Education.
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35, 128–152.
- Corley, K. G., & Gioia, D. A. (2004). Identity ambiguity and change in the wake of a corporate spin-off. *Administrative Science Quarterly*, 49, 173–208.
- Coupland, C., & Brown, A. D. (2004). Constructing organizational identities on the web: A case study of Royal Dutch/Shell. *Journal of Management Studies*, 41, 1325–1347.
- Courpasson, D., & Dany, F. (2003). Indifference or obedience? Business firms as democratic hybrids. *Organization Studies*, 24, 1231–1260.

- Denison, D. R. (1996). What is the difference between organizational culture and organizational climate? A native's point of view on a decade of paradigm wars. *Academy of Management Review*, 21, 619–654.
- Dingsøyr, T., Nerur, S., Balijepally, V., & Moe, N. B. (2012). A decade of agile methodologies: Towards explaining agile software development. *Journal of Systems and Software*, 85, 1213–1221.
- Dommermuth, W. P. (1989). *Promotion: Analysis, creativity, and strategy*. Boston: PWS-Kent Publishing Company.
- Dukerich, J. M., Kramer, R., & Parks, J. M. (1998). The dark side of organizational identification. In D. A. Whetten & P. C. Godfrey (Eds.), *Identity in organizations: Building theory through conversations* (pp. 245–256). Thousand Oaks, CA: SAGE Publications.
- Dutton, J. E., Dukerich, J. M., & Harquail, C. V. (1994). Organizational images and member identification. *Administrative Science Quarterly*, 39, 239–263.
- Dybå, T., & Dingsøyr, T. (2008). Empirical studies of agile software development: A systematic review. *Information and Software Technology*, 50, 833–859.
- Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review*, 14, 532–550.
- Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of Management Journal*, 50, 25–32.
- Elsbach, K. D., & Kramer, R. M. (1996). Members' responses to organizational identity threats: Encountering and countering the Business Week rankings. *Administrative Science Quarterly*, 41, 442–476.
- Essers, C., & Benschop, Y. (2007). Enterprising identities: Female entrepreneurs of Moroccan or Turkish origin in the Netherlands. *Organization Studies*, 28, 49–69.
- Ferraris, C. R., Carveth, R., & Parrish-Sprowl, J. (1993). Interface Precision Benchworks: A case study in organizational identification. *Journal of Applied Communication Research*, 21, 343–357.
- Fogel, A. (1993). *Developing through relationships*. Chicago: University of Chicago Press.
- Foreman, P. O., & Parent, M. M. (2008). The process of organizational identity construction in iterative organizations. *Corporate Reputation Review*, 11, 222–244.
- Fournier, V. (1999). The appeal to 'professionalism' as a disciplinary mechanism. *Sociological Review*, 47, 280–307.
- Fowler, M., & Highsmith, J. (2001). The agile manifesto. *Software Development*, 9(8), 28–35.
- Freeman, L. C. (1979). Centrality in social networks conceptual clarification. *Social Networks*, 1(3), 215–239.
- Freeman, R. E. (1983). Strategic management: A stakeholder approach. *Advances in Strategic Management*, 1(1), 31–60.
- Gabriel, Y. (1999). Beyond happy families: A critical reevaluation of the control-resistance-identity triangle. *Human Relations*, 52, 179–203.
- Galbraith, J. R. (1974). Organization design: An information processing view. *Interfaces*, 4(3), 28–36.
- Giddens, A. (1984). *The constitution of society: Outline of the theory of structuration*. Berkeley, CA: University of California Press.
- Gioia, D. A., Schultz, M., & Corley, K. G. (2000). Organizational identity, image, and adaptive instability. *Academy of Management Review*, 25, 63–81.
- Gioia, D. A., & Thomas, J. B. (1996). Identity, image, and issue interpretation: Sensemaking during strategic change in academia. *Administrative Science Quarterly*, 41, 370–403.
- Hopkins, T. K. (1964). *The exercise of influence in small groups*. Totowa, NJ: Bedminster.
- Jick, T. D. (1979). Mixing qualitative and quantitative methods: Triangulation in action. *Administrative Science Quarterly*, 24, 602–611.
- Johnson, R. E., & Yang, L. Q. (2010). Commitment and motivation at work: The relevance of employee identity and regulatory focus. *Academy of Management Review*, 35, 226–245.
- Kaplan, S. (1987). Aesthetics, affect, and cognition environmental preference from an evolutionary perspective. *Environment and Behavior*, 19, 3–32.
- Kärreman, D., & Alvesson, M. (2004). Cages in tandem: Management control, social identity, and identification in a knowledge-intensive firm. *Organization*, 11, 149–175.
- Kelly, J. R., Jackson, J. W., & Hutson-Comeaux, S. L. (1997). The effects of time pressure and task differences on influence modes and accuracy in decision-making groups. *Personality and Social Psychology Bulletin*, 23, 10–22.

- Kramer, R. M. (1991). Intergroup relations and organizational dilemmas: The role of categorization processes. In B. M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (pp. 191–228). Greenwich: JAI Press.
- Kumar, N., Stern, L. W., & Anderson, J. C. (1993). Conducting interorganizational research using key informants. *Academy of Management Journal*, 36, 1633–1651.
- Heckscher, C., & Donnellon, A. (1994). *The post-bureaucratic organization: New perspectives on organizational change*. London: SAGE Publications.
- Hill, C. W., & Jones, T. M. (1992). Stakeholder–agency theory. *Journal of Management Studies*, 29, 131–154.
- Hogg, M. A., & Abrams, D. (1988). *Social identification*. New York: Routledge, Chapman & Hall.
- Hogg, M. A., & Terry, D. I. (2000). Social identity and self-categorization processes in organizational contexts. *Academy of Management Review*, 25, 121–140.
- Langfield-Smith, K. (1997). Management control systems and strategy: A critical review. *Accounting, Organizations, and Society*, 22, 207–232.
- Larson, G. S., & Tompkins, P. K. (2005). Ambivalence and resistance: A study of management in a concertive control system. *Communication Monographs*, 72, 1–21.
- Leary, M. R. (1983). *Understanding social anxiety: Social, personality and clinical perspectives*. London: SAGE Publications.
- Lee, G., & Xia, W. (2010). Toward agile: An integrated analysis of quantitative and qualitative field data. *Management Information Systems Quarterly*, 34, 87–114.
- Lord, R. G., & Hanges, P. J. (1987). A control system model of organizational motivation: Theoretical development and applied implications. *Behavioral Science*, 32(3), 161–178.
- Martin, R. C. (2003). *Agile software development: Principles, patterns, and practices*. Upper Saddle River, NJ: Prentice Hall.
- McCaslin, M. (2009). Co-regulation of student motivation and emergent identity. *Educational Psychologist*, 44, 137–146.
- Meyer, J. W., & Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology*, 83, 340–363.
- Mitchell, R. K., Agle, B. R., & Wood, D. J. (1997). Toward a theory of stakeholder identification and salience: Defining the principle of who and what really counts. *Academy of Management Review*, 22, 853–886.
- Nerur, S., & Balijepally, V. (2007). Theoretical reflections on agile development methodologies. *Communications of the ACM*, 50, 79–83.
- Owen-Smith, J., & Powell, W. W. (2008). Networks and institutions. In R. Greenwood, C. Oliver, R. Suddaby, & K. Sahlin-Andersson (Eds.), *The Sage handbook of organizational institutionalism* (pp. 596–623). London: SAGE Publications.
- Pfeffer, J. (1981). *Power in organizations*. Marshfield, MA: Pitman.
- Podd, M. H. (1972). Ego identity status and morality: The relationship between two developmental constructs. *Developmental Psychology*, 6, 497–507.
- Postmes, T., & Jetten, J. (2006). *Individuality and the group: Advances in social identity*. London: SAGE Publications.
- Rowley, T. J. (1997). Moving beyond dyadic ties: A network theory of stakeholder influences. *Academy of Management Review*, 22, 887–910.
- Schein, E. H. (1990) Organizational culture. *American Psychologist*, 45, 109–119.
- Schlenker, B. R. (1985) Identities, identifications, and relationships. In V. Derlega (Ed.), *Communication, intimacy and close relationships* (pp. 65–99). New York: Academic Press.
- Schlenker, B. R., & Leary, M. R. (1982). Social anxiety and self-presentation: A conceptualization model. *Psychological Bulletin*, 92, 641–669.
- Schwaber, K. (1997). Scrum development process. In J. V. Sutherland, D. Patel, C. Casanave, J. Miller, & G. Hollowell. (Eds.), *Business object design and implementation* (pp. 117–134). London: Springer.
- Schwaber, K., & Beedle, M. (2002). *Agile software development with scrum*. Upper Saddle River, NJ: Prentice-Hall.
- Scott, S. G., & Lane, V. R. (2000). A stakeholder approach to organizational identity. *Academy of Management Review*, 25, 43–62.

- Scott, W. R. (1995). *Institutions and organizations*. Thousand Oaks, CA: SAGE Publications.
- Sewell, G. (1998). The discipline of teams: The control of team-based industrial work through electronic and peer surveillance. *Administrative Science Quarterly*, 43, 397–428.
- Shamir, B. (1991). Meaning, self and motivation in organizations. *Organization Studies*, 12, 405–424.
- Sluss, D. M., & Ashforth, B. E. (2007). Relational identity and identification: Defining ourselves through work relationships. *Academy of Management Review*, 32, 9–32.
- Strauss, A. L. (1978). *Negotiations: Varieties, contexts, processes, and social order*. San Francisco: Jossey-Bass.
- Suchman, M. C. (1995). Managing legitimacy: Strategic and institutional approaches. *Academy of Management Journal*, 20, 571–610.
- Sutherland, J., & Schwaber, K. (2007). *The scrum papers: Nuts, bolts, and origins of an agile method*. Cambridge, MA: Scrum, Inc.
- Sveningsson, S., & Alvesson, M. (2003). Managing managerial identities: Organizational fragmentation, discourse and identity struggle. *Human Relations*, 56, 1163–1193.
- Swidler, A. (1986). Culture in action: Symbols and strategies. *American Sociological Review*, 51, 273–286.
- Takeuchi, H., & Nonaka, I. (1986). The new product development game. *Harvard Business Review*, 64, 137–146.
- Thornton, P. H. (2002). The rise of the corporation in a craft industry: Conflict and conformity in institutional logics. *Academy of Management Journal*, 45, 81–101.
- Tompkins, P. K., & Cheney, G. (1985). Communication and unobtrusive control in contemporary organizations. In R. D. McPhee & P. K. Tompkins (Eds.), *Organizational communication: Traditional themes and new directions* (pp. 179–210). London: SAGE Publications.
- Turner, J. C. (1982). Towards a cognitive redefinition of the social group. *Current Psychology of Cognition*, 1(2), 93–118.
- Van Maanen, J., & Schein, E. H. (1979). Toward a theory of organizational socialization. In B. Staw & L. Cummings (Eds.), *Research in organizational behavior*. Greenwich, CT: JAI Press.
- Wasserman, S., & Galaskiewicz, J. (1994). *Advances in social network analysis: Research in the social and behavioral sciences*. London: SAGE Publications.
- White, H. C. (1992). *Identity and control: A structural theory of social action*. Princeton, NJ: Princeton University Press.
- Zanoni, P., & Janssens, M. (2007). Minority employees engaging with (diversity) management: An analysis of control, agency, and microemancipation. *Journal of Management Studies*, 44, 1371–1397.

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